

Nature | Volume 657 | Issue 8132

Issue date: 2026-09-17 | Retrieved: 2026-09-17T00:36:47.688247+00:00 | 15 relevant primary-research papers

Evidence discipline

Paper-supported findings, published enabling evidence, cross-species inference, speculative hypotheses, and proposed pilot experiments are labeled separately. Access notes state when only an abstract or limited preview was available.

Screening notes

- Official publisher issue page was reachable and contained the detected volume and issue identifiers.
- Screened 15 biology-relevant journal articles after excluding reviews, perspectives, news, corrections, and editorials by metadata/title rules.
- All summaries are deterministic extractions from the bounded public evidence packet; no language-model API or paid research service was used.

Official issue

<https://www.nature.com/nature/volumes/657/issues/8132>

Why Xenopus now

Early Xenopus combines rapid external development, targeted microinjection, explants, lineage tracing, live imaging, F0 CRISPR, rescue, and scalable replication. It can convert abstract-supported candidate mechanisms into spatially resolved causal tests before committing to slower mammalian models.

Aneuploidy selects for the acquisition of driver genes in breast cancer

Khalid N. Al-Zahrani, Ellen R. Langille, Jocelyn Nurtanto, Andreea Obersterescu, Katie Teng, Christopher Lowden, Julien Dessapt, Cynthia H. Chiu, et al.

journal-article | Nature 657(8132) | 2026-09-17

DOI 10.1038/s41586-026-10752-9

Access: Abstract retrieved from Europe PMC; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	Chromosome instability is highly prevalent in cancer and drives large-scale chromosomal imbalances, known as aneuploidies 1-4 .
WHY IT MATTERS	The study advances mechanistic understanding of Aneuploidy selects for the acquisition of driver genes in breast cancer and creates a defined basis for testing conservation and developmental function in a vertebrate embryo.
MAIN / NEW METHODS	• CRISPR genome editing • Genetic perturbation and rescue
KEY EVIDENCE	• Chromosome instability is highly prevalent in cancer and drives large-scale chromosomal imbalances, known as aneuploidies 1-4 . • Here we established a CRISPR knockout- and activation-linked assay (CRISPR-KOALA), enabling high-throughput bidirectional genetic screens in immunocompetent mouse models of cancer. • Mechanistically, we identify that PLGRKT is a potent oncogene that lies on chromosome 9p and show that its tumour-promoting activity is associated with highly stress-resistant mitochondria and an increased ability to detoxify reactive oxygen species.
LIMITATIONS	• This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. • The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. • The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of Aneuploidy selects for the acquisition of driver genes in breast cancer

IMPORTANCE / SIGNIFICANCE

Gene Regulation And Chromatin is central to how embryonic cells convert molecular state into reproducible tissue organization. Establishing whether the featured mechanism operates during early vertebrate development could reveal a conserved control point and a tractable route to explain developmental variability without overstating clinical translation.

KNOWLEDGE GAP

The featured paper does not establish whether the focal regulator or mechanism described by the featured paper acts during early vertebrate development, whether its effect is cell-autonomous, which developmental window is sensitive, or whether molecular and morphogenetic phenotypes can be rescued independently.

RATIONALE AND XENOPUS ADVANTAGE

The paper supplies a candidate mechanism, while early *Xenopus* provides rapid external development, targeted blastomere injection, explants, lineage tracing, live imaging, inexpensive biological replication across clutches, and direct comparison of molecular and tissue-scale outcomes.

CENTRAL HYPOTHESIS

Speculative: the focal regulator or mechanism described by the featured paper acts cell-autonomously during stages 8-14 to stabilize pathway activity, lineage allocation, and spatial marker-domain size; acute loss will reduce the predicted molecular readout and increase pattern variance, while stage-matched wild-type rescue will restore both.

EXPERIMENTAL DESIGN

Use *Xenopus tropicalis* for F0 genetics and *Xenopus laevis* explants. Target animal-cap, organizer, and mesendodermal lineages by injecting two independent CRISPR RNPs at the 4- to 8-cell stage with a fluorescent lineage tracer; analyze stages 8-14. Groups: uninjected, non-targeting RNP, each guide alone, pooled guides, dose series, wild-type mRNA rescue, perturbation-resistant rescue, and temporally delayed perturbation where a validated small molecule or inducible construct exists. Primary readouts: pathway activity, lineage allocation, and spatial marker-domain size. Secondary readouts: cell behavior, developmental timing, and gastrulation morphology, amplicon sequencing, RT-qPCR or in situ hybridization, and an orthogonal biochemical or reporter assay. Use at least three independent spawnings, 20-30 embryos per group per spawning for morphology, and 8-10 pooled or individually profiled samples per molecular condition; randomize embryos, blind image scoring, and analyze clutch as a random effect with multiplicity-controlled planned contrasts. Decision point: advance to epistasis and spatial single-cell profiling only if both guides reproduce the effect and rescue restores at least 50% of the molecular and developmental phenotype.

EXPECTED RESULTS AND INTERPRETATIONS

Support would be a guide-concordant loss of the predicted molecular readout, increased spatial or embryo-to-embryo variance, and rescue of both molecular and tissue phenotypes. This would place the mechanism upstream of developmental robustness. A molecular phenotype without morphogenesis would suggest buffering or a parallel tissue effector and would redirect the next aim to sensitized or explant conditions. A morphogenetic phenotype without the predicted reporter change would argue for an alternative biochemical function. No reproducible phenotype after confirmed editing would refute an obligate early role and shift the model toward redundancy, later function, or species-specific deployment.

POTENTIAL PITFALLS

- F0 mosaicism may weaken or spatially blur phenotypes.
- Paralog redundancy or maternal stores may mask an early requirement.

- High-dose perturbation may cause nonspecific toxicity or developmental delay.
- The featured mechanism may not be conserved or may use different effectors in amphibians.
- Abstract-only evidence may omit critical context needed to choose the most specific assay.

ALTERNATIVE STRATEGIES

- Map editing per embryo and enrich analysis for high-editing specimens; establish stable alleles if mosaicism remains limiting.
- Target expressed paralogs jointly, use maternal-zygotic strategies where feasible, or test dominant-negative and degron-based acute perturbations.
- Use lower doses, later induction, tissue-restricted injection, explants, and matched survival/staging controls to separate mechanism from toxicity.
- Test *Xenopus laevis* and *X. tropicalis* orthologs, then validate the key interaction biochemically or in a complementary cell model.
- Before full-scale work, manually review the linked paper's figures and methods and revise the reporter or perturbation to match the demonstrated mechanism.

PUBLISHED ENABLING EVIDENCE AND PROPOSED PRELIMINARY-DATA PLAN

Published enabling evidence and proposed preliminary-data plan: no unpublished user-laboratory data are assumed. The featured paper is the cited cross-species enabling source; direct *Xenopus* evidence was not automatically verified. Pilot 1: map candidate expression or pathway activity at 5-6 early stages using RT-qPCR and in situ hybridization (three spawnings; expected Figure 1A-B); go if a reproducible spatial or temporal signal exceeds twofold over the lowest stage. Pilot 2: test two CRISPR RNPs plus non-targeting control in three spawnings with at least 25 embryos per group, amplicon sequencing, blinded morphology, and the primary molecular readout (Figure 1C-E); go if both guides produce the same directional effect of at least 20% with acceptable survival. Pilot 3: perform wild-type mRNA rescue in three spawnings with at least 20 embryos per group (Figure 1F-G); go if rescue reverses at least 50% of the molecular and phenotype effect. Failure at any threshold triggers the paired alternative strategy rather than a claim of preliminary support.

NOVELTY medium-high - transfers a recent mechanism into a spatially and temporally tractable vertebrate embryo

FEASIBILITY high for the pilots - injection, explants, imaging, genotyping, and rescue are routine *Xenopus* capabilities

RISK medium - conservation, maternal contribution, and mosaicism are explicit decision points

GRANT FIT

Exploratory Aim 1 for conservation and causality; successful pilots support Aim 2 epistasis and Aim 3 spatial mechanism.

STRONGEST REVIEWER-FACING PREMISE

Xenopus can test when, where, and whether a newly reported cellular mechanism controls intact vertebrate development in one experimentally integrated system.

SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\) 10.1038/s41586-026-10752-9](#)

Prior therapy defines mutation profiles in childhood cancer at relapse

Mehdi Layeghifard, Marcos Díaz-Gay, Erik N. Bergstrom, Mathepan J. Mahendralingam, Nicholas Light, Sasha Blay, Joshua O. Nash, Nathaniel D. Anderson, et al.

journal-article | Nature 657(8132) | 2026-09-17

DOI 10.1038/s41586-026-10803-1

Access: Abstract retrieved from Europe PMC; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	These data highlight opportunities for treatment de-escalation and the future possibility of tracking resistant clones before expansion.
WHY IT MATTERS	This work provides genomic evidence for the critical mutagenic effects of chemotherapy in childhood cancer, as a specific driver of tumour evolution.
MAIN / NEW METHODS	• Publisher-deposited title/abstract analysis; consult the linked paper for complete technical details
KEY EVIDENCE	• The retrieved publisher-deposited metadata identifies this item as a journal article in the detected volume and issue. • The DOI provides the canonical path for manual verification of figures and full methods.
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SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\) 10.1038/s41586-026-10803-1](https://doi.org/10.1038/s41586-026-10803-1)

A compendium of next-generation patient-derived models for diverse cancers

Dina ElHarouni, Mushriq Al-Jazrawe, Seongmin Choi, Merve Dede, Toshinori Hinoue, Sean A. Misek, Heeju Noh, Luca Zanella, et al.

journal-article | Nature 657(8132) | 2026-09-17

DOI 10.1038/s41586-026-10806-y

Access: Publisher-deposited abstract retrieved through Crossref; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	Analyses of 421 matched tumour-model pairs reveal high genetic (97.8%) and epigenetic (95%) concordance and define correlates of model discordance.
WHY IT MATTERS	Finally, we characterize model preservation of extrachromosomal DNA and post-treatment mutational signatures to provide opportunities to study therapeutic resistance.
MAIN / NEW METHODS	• Single-nucleus multi-omics • Organoid or stem-cell model • Biochemical reconstitution
KEY EVIDENCE	• Analyses of 421 matched tumour-model pairs reveal high genetic (97.8%) and epigenetic (95%) concordance and define correlates of model discordance. • Single-nucleus RNA sequencing of tumour-model pairs reveals subsets of models in which culture conditions significantly influence cell states.
LIMITATIONS	• This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. • The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. • The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of A compendium of next-generation patient-derived models for diverse cancers

IMPORTANCE / SIGNIFICANCE

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A tumour-derived organoid biobank maps cancer gene dependencies

C. Herranz-Ors, S. G. Bhosle, A. E. Beck, J. G. R. Gilbert, G. Picco, J. Espejo Valle-Inclan, F. Muyas, S. Valentini, et al.

journal-article | Nature 657(8132) | 2026-09-17

DOI 10.1038/s41586-026-10830-y

Access: Publisher-deposited abstract retrieved through Crossref; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	This open, publicly available resource provides a systematic map of gene dependencies in patient-derived organoids, expanding the model diversity and mechanistic insight needed to advance precision oncology.
WHY IT MATTERS	This open, publicly available resource provides a systematic map of gene dependencies in patient-derived organoids, expanding the model diversity and mechanistic insight needed to advance precision oncology.
MAIN / NEW METHODS	• CRISPR genome editing • Organoid or stem-cell model
KEY EVIDENCE	• Abstract Cancer cell lines remain foundational for research and drug discovery, yet they incompletely capture tumour diversity, lack linked patient context, and have undergone adaptation to culture. • Integrative analyses revealed genomic and clinical markers of dependency across common and rare subtypes, identified organoid-specific essential genes, and revealed targetable vulnerabilities following tumour evolution in paired pre- and post-treatment samples.
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Structural basis of the RNA-editing cascade in trypanosome mitochondria

Yun-Tao Liu, Andres F. Vacas, Jonathan Jih, Xiaojing Zhao, Clinton Yu, Jane K. J. Lee, Takuma Suematsu, Md Solayman, et al.

journal-article | Nature 657(8132) | 2026-09-17

DOI 10.1038/s41586-026-10831-x

Access: Abstract retrieved from Europe PMC; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	Resembling dragonflies, with a head, thorax-like core, tail and wings, these ribonucleoproteins mediate the uridine deletion and uridine insertion cascades, respectively.
WHY IT MATTERS	The study advances mechanistic understanding of Structural basis of the RNA-editing cascade in trypanosome mitochondria and creates a defined basis for testing conservation and developmental function in a vertebrate embryo.
MAIN / NEW METHODS	• Electron microscopy • Quantitative imaging
KEY EVIDENCE	• Resembling dragonflies, with a head, thorax-like core, tail and wings, these ribonucleoproteins mediate the uridine deletion and uridine insertion cascades, respectively. • Together, these structures reveal how gRNA-directed substrate recognition, mRNA cleavage, uridine deletion and insertion and ligation are integrated in a single macromolecular machine.
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Speculative: the focal regulator or mechanism described by the featured paper acts cell-autonomously during stages 8-14 to stabilize pathway activity, lineage allocation, and spatial marker-domain size; acute loss will reduce the predicted molecular readout and increase pattern variance, while stage-matched wild-type rescue will restore both.

EXPERIMENTAL DESIGN

Use *Xenopus tropicalis* for F0 genetics and *Xenopus laevis* explants. Target animal-cap, organizer, and mesendodermal lineages by injecting two independent CRISPR RNPs at the 4- to 8-cell stage with a fluorescent lineage tracer; analyze stages 8-14. Groups: uninjected, non-targeting RNP, each guide alone, pooled guides, dose series, wild-type mRNA rescue, perturbation-resistant rescue, and temporally delayed perturbation where a validated small molecule or inducible construct exists. Primary readouts: pathway activity, lineage allocation, and spatial marker-domain size. Secondary readouts: cell behavior, developmental timing, and gastrulation morphology, amplicon sequencing, RT-qPCR or in situ hybridization, and an orthogonal biochemical or reporter assay. Use at least three independent spawnings, 20-30 embryos per group per spawning for morphology, and 8-10 pooled or individually profiled samples per molecular condition; randomize embryos, blind image scoring, and analyze clutch as a random effect with multiplicity-controlled planned contrasts. Decision point: advance to epistasis and spatial single-cell profiling only if both guides reproduce the effect and rescue restores at least 50% of the molecular and developmental phenotype.

EXPECTED RESULTS AND INTERPRETATIONS

Support would be a guide-concordant loss of the predicted molecular readout, increased spatial or embryo-to-embryo variance, and rescue of both molecular and tissue phenotypes. This would place the mechanism upstream of developmental robustness. A molecular phenotype without morphogenesis would suggest buffering or a parallel tissue effector and would redirect the next aim to sensitized or explant conditions. A morphogenetic phenotype without the predicted reporter change would argue for an alternative biochemical function. No reproducible phenotype after confirmed editing would refute an obligate early role and shift the model toward redundancy, later function, or species-specific deployment.

POTENTIAL PITFALLS

- F0 mosaicism may weaken or spatially blur phenotypes.
- Paralog redundancy or maternal stores may mask an early requirement.

- High-dose perturbation may cause nonspecific toxicity or developmental delay.
- The featured mechanism may not be conserved or may use different effectors in amphibians.
- Abstract-only evidence may omit critical context needed to choose the most specific assay.

ALTERNATIVE STRATEGIES

- Map editing per embryo and enrich analysis for high-editing specimens; establish stable alleles if mosaicism remains limiting.
- Target expressed paralogs jointly, use maternal-zygotic strategies where feasible, or test dominant-negative and degron-based acute perturbations.
- Use lower doses, later induction, tissue-restricted injection, explants, and matched survival/staging controls to separate mechanism from toxicity.
- Test *Xenopus laevis* and *X. tropicalis* orthologs, then validate the key interaction biochemically or in a complementary cell model.
- Before full-scale work, manually review the linked paper's figures and methods and revise the reporter or perturbation to match the demonstrated mechanism.

PUBLISHED ENABLING EVIDENCE AND PROPOSED PRELIMINARY-DATA PLAN

Published enabling evidence and proposed preliminary-data plan: no unpublished user-laboratory data are assumed. The featured paper is the cited cross-species enabling source; direct *Xenopus* evidence was not automatically verified. Pilot 1: map candidate expression or pathway activity at 5-6 early stages using RT-qPCR and in situ hybridization (three spawnings; expected Figure 1A-B); go if a reproducible spatial or temporal signal exceeds twofold over the lowest stage. Pilot 2: test two CRISPR RNPs plus non-targeting control in three spawnings with at least 25 embryos per group, amplicon sequencing, blinded morphology, and the primary molecular readout (Figure 1C-E); go if both guides produce the same directional effect of at least 20% with acceptable survival. Pilot 3: perform wild-type mRNA rescue in three spawnings with at least 20 embryos per group (Figure 1F-G); go if rescue reverses at least 50% of the molecular and phenotype effect. Failure at any threshold triggers the paired alternative strategy rather than a claim of preliminary support.

NOVELTY medium-high - transfers a recent mechanism into a spatially and temporally tractable vertebrate embryo	FEASIBILITY high for the pilots - injection, explants, imaging, genotyping, and rescue are routine <i>Xenopus</i> capabilities	RISK medium - conservation, maternal contribution, and mosaicism are explicit decision points
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GRANT FIT

Exploratory Aim 1 for conservation and causality; successful pilots support Aim 2 epistasis and Aim 3 spatial mechanism.

STRONGEST REVIEWER-FACING PREMISE

Xenopus can test when, where, and whether a newly reported cellular mechanism controls intact vertebrate development in one experimentally integrated system.

SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\) 10.1038/s41586-026-10831-x](#)

A dependency map enhanced with next-generation 3D cancer models

James V. Neiswender, Samuel Maffa, Lisa Brenan, Dina ElHarouni, Yejie Yun, Isabella Boyle, Kirsty Wienand, Haider Inam, et al.

journal-article | Nature 657(8132) | 2026-09-17

DOI 10.1038/s41586-026-10843-7

Access: Abstract retrieved from Europe PMC; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	This strategy enables the expansion of DepMap to cover new genomic and molecular subtypes and to identify new biomarker-associated vulnerabilities.
WHY IT MATTERS	Despite advances in precision oncology, effective personalized treatments are still lacking for most patients with cancer 1 .
MAIN / NEW METHODS	• Single-nucleus multi-omics • CRISPR genome editing • Organoid or stem-cell model • Genetic perturbation and rescue
KEY EVIDENCE	• The Cancer Dependency Map (DepMap) accelerates this field by systematically identifying cancer vulnerabilities in diverse preclinical models. • Data from over 1,300 cell lines have led to the discovery of new therapeutic strategies across multiple tumour types 2 . • This strategy enables the expansion of DepMap to cover new genomic and molecular subtypes and to identify new biomarker-associated vulnerabilities.
LIMITATIONS	• This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. • The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. • The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of A dependency map enhanced with next-generation 3D cancer models

IMPORTANCE / SIGNIFICANCE

Gene Regulation And Chromatin is central to how embryonic cells convert molecular state into reproducible tissue organization. Establishing whether the featured mechanism operates during early vertebrate development could reveal a conserved control point and a tractable route to explain developmental variability without overstating clinical translation.

KNOWLEDGE GAP

The featured paper does not establish whether the focal regulator or mechanism described by the featured paper acts during early vertebrate development, whether its effect is cell-autonomous, which developmental window is sensitive, or whether molecular and morphogenetic phenotypes can be rescued independently.

RATIONALE AND XENOPUS ADVANTAGE

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EXPERIMENTAL DESIGN

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GRANT FIT

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STRONGEST REVIEWER-FACING PREMISE

Xenopus can test when, where, and whether a newly reported cellular mechanism controls intact vertebrate development in one experimentally integrated system.

SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\) 10.1038/s41586-026-10843-7](https://doi.org/10.1038/s41586-026-10843-7)

Diverse bacterial pattern recognition receptors sense the core phage proteome

Hyunbin Lee, Sofia Luengo-Woods, Jianxiu Zhang, Kira S. Makarova, Yuri I. Wolf, Collin Chiu, Simone A. Evans, Junyi Chen, et al.

journal-article | Nature 657(8132) | 2026-09-17

DOI 10.1038/s41586-026-10852-6

Access: Publisher-deposited abstract retrieved through Crossref; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	Here we conduct a systematic phylogenetic analysis of prokaryotic STAND NTPases and identify at least 90 structurally distinct families associated with antiviral defence.
WHY IT MATTERS	The study advances mechanistic understanding of Diverse bacterial pattern recognition receptors sense the core phage proteome and creates a defined basis for testing conservation and developmental function in a vertebrate embryo.
MAIN / NEW METHODS	• Electron microscopy • Quantitative imaging
KEY EVIDENCE	• Here we conduct a systematic phylogenetic analysis of prokaryotic STAND NTPases and identify at least 90 structurally distinct families associated with antiviral defence. • We first show that the uncharacterized Avs7 family recognizes the major capsid protein (MCP) of tailed phages. • Three cryogenic electron microscopy structures of Salmonella enterica Avs7 reveal an asymmetric, butterfly-shaped tetramer that assembles stepwise through large, MCP-induced conformational changes, incorporating bacterial elongation factor Tu (EF-Tu) as a structural component that enhances defence.
LIMITATIONS	• This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. • The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. • The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of Diverse bacterial pattern recognition receptors sense the core phage proteome

IMPORTANCE / SIGNIFICANCE

Immunity And Host Response is central to how embryonic cells convert molecular state into reproducible tissue organization. Establishing whether the featured mechanism operates during early vertebrate development could reveal a conserved control point and a tractable route to explain developmental variability without overstating clinical translation.

KNOWLEDGE GAP

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RATIONALE AND XENOPUS ADVANTAGE

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CENTRAL HYPOTHESIS

Speculative: the focal regulator or mechanism described by the featured paper acts cell-autonomously during stages 8-18 to stabilize cell-stress signaling, epithelial integrity, and lineage-specific transcription; acute loss will reduce the predicted molecular readout and increase pattern variance, while stage-matched wild-type rescue will restore both.

EXPERIMENTAL DESIGN

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ALTERNATIVE STRATEGIES

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NOVELTY medium-high - transfers a recent mechanism into a spatially and temporally tractable vertebrate embryo	FEASIBILITY high for the pilots - injection, explants, imaging, genotyping, and rescue are routine <i>Xenopus</i> capabilities	RISK medium - conservation, maternal contribution, and mosaicism are explicit decision points
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GRANT FIT

Exploratory Aim 1 for conservation and causality; successful pilots support Aim 2 epistasis and Aim 3 spatial mechanism.

STRONGEST REVIEWER-FACING PREMISE

Xenopus can test when, where, and whether a newly reported cellular mechanism controls intact vertebrate development in one experimentally integrated system.

SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\) 10.1038/s41586-026-10852-6](#)

Antigen presentation by CD40+MHC-II+ astrocytes promotes CNS autoimmunity

Joon-Hyuk Lee, Zhaorong Li, Joselyn S. Soto, Ah-Ram Kim, Tomer Illouz, Carolina M. Polonio, Michael Kilian, Jessica E. Kenison, et al.

journal-article | Nature 657(8132) | 2026-09-17

DOI 10.1038/s41586-026-10860-6

Access: Abstract retrieved from Europe PMC; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	In summary, these studies define a previously unrecognized mechanism by which astrocytes promote CNS autoimmunity.
WHY IT MATTERS	Using in vivo subproteomic approaches 4 and AlphaFold-Multimer predictions 5 , we established that CD40 activation in astrocytes by CD40L expressed by CD4 + T cells induces the accumulation of PLIN4-positive lipid droplets, which provide acetyl-CoA to promote p65 acetylation-dependent NF-kappaB activation and antigen presentation.
MAIN / NEW METHODS	• Single-cell transcriptomics • Single-nucleus multi-omics • CRISPR genome editing • Proteomics or mass spectrometry • Genetic perturbation and rescue • Biochemical reconstitution
KEY EVIDENCE	• Here using rabies barcode interaction detection followed by sequencing 2 , in combination with single-cell RNA sequencing, in vitro co-culture systems and cell-specific in vivo CRISPR-Cas9-based genetic perturbation studies, we established that astrocytes expressing CD40 and MHC-II promote CNS T cell autoimmunity. • We harnessed universal labelling immune partnerships by SorTagging intercellular contacts 3 to analyse astrocyte-interacting CD4 + T cells, finding that direct astrocyte-CD4 + T cell interactions enhance pathogenic T helper 17 cell responses in experimental autoimmune encephalomyelitis. • Using in vivo subproteomic approaches 4 and AlphaFold-Multimer predictions 5 , we established that CD40 activation in astrocytes by CD40L expressed by CD4 + T cells induces the accumulation of PLIN4-positive lipid droplets, which provide acetyl-CoA to promote p65 acetylation-dependent NF-kappaB activation and antigen presentation.
LIMITATIONS	• This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. • The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. • The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of Antigen presentation by CD40+MHC-II+ astrocytes promotes CNS autoimmunity

IMPORTANCE / SIGNIFICANCE

Immunity And Host Response is central to how embryonic cells convert molecular state into reproducible tissue organization. Establishing whether the featured mechanism operates during early vertebrate development could reveal a conserved control point and a tractable route to explain developmental variability without overstating clinical translation.

KNOWLEDGE GAP

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RATIONALE AND XENOPUS ADVANTAGE

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CENTRAL HYPOTHESIS

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EXPERIMENTAL DESIGN

Use *Xenopus tropicalis* for F0 genetics and *Xenopus laevis* explants. Target epidermal and mesendodermal lineages before mature immunity by injecting two independent CRISPR RNPs at the 4- to 8-cell stage with a fluorescent lineage tracer; analyze stages 8-18. Groups: uninjected, non-targeting RNP, each guide alone, pooled guides, dose series, wild-type mRNA rescue, perturbation-resistant rescue, and temporally delayed perturbation where a validated small molecule or inducible construct exists. Primary readouts: cell-stress signaling, epithelial integrity, and lineage-specific transcription. Secondary readouts: survival, barrier morphology, and developmental timing, amplicon sequencing, RT-qPCR or in situ hybridization, and an orthogonal biochemical or reporter assay. Use at least three independent spawnings, 20-30 embryos per group per spawning for morphology, and 8-10 pooled or individually profiled samples per molecular condition; randomize embryos, blind image scoring, and analyze clutch as a random effect with multiplicity-controlled planned contrasts. Decision point: advance to epistasis and spatial single-cell profiling only if both guides reproduce the effect and rescue restores at least 50% of the molecular and developmental phenotype.

EXPECTED RESULTS AND INTERPRETATIONS

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POTENTIAL PITFALLS

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STRONGEST REVIEWER-FACING PREMISE

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SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\)](#) 10.1038/s41586-026-10860-6

FNIP1 variants are associated with favourable metabolism in 1 million humans

George Hindy, Rene C. Adam, Olukayode Sosina, Dwaine Pryce, David Blair, Joseph Herman, Joseph Lee, Peter Dornbos, et al.

journal-article Nature 657(8132) 2026-09-17	
DOI 10.1038/s41586-026-10864-2	
Access: Abstract retrieved from Europe PMC; no full-text review is claimed. Evidence status: abstract-supported	
MAIN DISCOVERY	We identify 59 independent genes (P -7) that are enriched for liver- and adipose-expressed master regulators of energy balance, storage and metabolism; 23 (39%) of these genes encode approved or clinical-stage drug targets.
WHY IT MATTERS	Our study implicates the FNIP1 pathway in human energy metabolism and highlights its inhibition as a potential therapeutic strategy in cardiometabolic disease.
MAIN / NEW METHODS	Genetic perturbation and rescue
KEY EVIDENCE	We identify 59 independent genes (P -7) that are enriched for liver- and adipose-expressed master regulators of energy balance, storage and metabolism; 23 (39%) of these genes encode approved or clinical-stage drug targets.
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Developmental timing and tissue autonomy of FNIP1 variants are associated with favourable metabolism in 1 million humans

IMPORTANCE / SIGNIFICANCE

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CENTRAL HYPOTHESIS

Speculative: the focal regulator or mechanism described by the featured paper acts cell-autonomously during stages 8-13 to stabilize organelle state, metabolite-sensitive reporters, and lineage-marker induction; acute loss will reduce the predicted molecular readout and increase pattern variance, while stage-matched wild-type rescue will restore both.

EXPERIMENTAL DESIGN

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- Before full-scale work, manually review the linked paper's figures and methods and revise the reporter or perturbation to match the demonstrated mechanism.

PUBLISHED ENABLING EVIDENCE AND PROPOSED PRELIMINARY-DATA PLAN

Published enabling evidence and proposed preliminary-data plan: no unpublished user-laboratory data are assumed. The featured paper is the cited cross-species enabling source; direct *Xenopus* evidence was not automatically verified. Pilot 1: map candidate expression or pathway activity at 5-6 early stages using RT-qPCR and in situ hybridization (three spawnings; expected Figure 1A-B); go if a reproducible spatial or temporal signal exceeds twofold over the lowest stage. Pilot 2: test two CRISPR RNPs plus non-targeting control in three spawnings with at least 25 embryos per group, amplicon sequencing, blinded morphology, and the primary molecular readout (Figure 1C-E); go if both guides produce the same directional effect of at least 20% with acceptable survival. Pilot 3: perform wild-type mRNA rescue in three spawnings with at least 20 embryos per group (Figure 1F-G); go if rescue reverses at least 50% of the molecular and phenotype effect. Failure at any threshold triggers the paired alternative strategy rather than a claim of preliminary support.

NOVELTY medium-high - transfers a recent mechanism into a spatially and temporally tractable vertebrate embryo	FEASIBILITY high for the pilots - injection, explants, imaging, genotyping, and rescue are routine <i>Xenopus</i> capabilities	RISK medium - conservation, maternal contribution, and mosaicism are explicit decision points
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GRANT FIT

Exploratory Aim 1 for conservation and causality; successful pilots support Aim 2 epistasis and Aim 3 spatial mechanism.

STRONGEST REVIEWER-FACING PREMISE

Xenopus can test when, where, and whether a newly reported cellular mechanism controls intact vertebrate development in one experimentally integrated system.

SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\) 10.1038/s41586-026-10864-2](#)

Human brain organoids record the passage of time over multiple years

Irene Faravelli, Noelia Antón-Bolaños, Anqi Wei, Tyler Faits, Abhishek Sampath Kumar, Sophia Andreadis, Rahel Kastli, Marta Montero Crespo, et al.

journal-article | Nature 657(8132) | 2026-09-17

DOI 10.1038/s41586-026-10877-x

Access: Abstract retrieved from Europe PMC; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	While human brain organoids offer a promising in vitro model, they have thus far been shown to largely mimic early stages of brain development.
WHY IT MATTERS	The study advances mechanistic understanding of Human brain organoids record the passage of time over multiple years and creates a defined basis for testing conservation and developmental function in a vertebrate embryo.
MAIN / NEW METHODS	• Organoid or stem-cell model • Biochemical reconstitution
KEY EVIDENCE	• While human brain organoids offer a promising in vitro model, they have thus far been shown to largely mimic early stages of brain development. • Using maturation-associated modules derived from endogenous human brain, we show that brain organoids transcriptionally age with cell type specificity over years in culture. • Whole-genome methylation profiling reveals that the predicted epigenomic age of organoids correlates precisely with time spent in vitro, and parallels epigenomic ageing in vivo.
LIMITATIONS	• This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. • The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. • The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of Human brain organoids record the passage of time over multiple years

IMPORTANCE / SIGNIFICANCE

Neural Development And Function is central to how embryonic cells convert molecular state into reproducible tissue organization. Establishing whether the featured mechanism operates during early vertebrate development could reveal a conserved control point and a tractable route to explain developmental variability without overstating clinical translation.

KNOWLEDGE GAP

The featured paper does not establish whether the focal regulator or mechanism described by the featured paper acts during early vertebrate development, whether its effect is cell-autonomous, which developmental window is sensitive, or whether molecular and morphogenetic phenotypes can be rescued independently.

RATIONALE AND XENOPUS ADVANTAGE

The paper supplies a candidate mechanism, while early *Xenopus* provides rapid external development, targeted blastomere injection, explants, lineage tracing, live imaging, inexpensive biological replication across clutches, and direct comparison of molecular and tissue-scale outcomes.

CENTRAL HYPOTHESIS

Speculative: the focal regulator or mechanism described by the featured paper acts cell-autonomously during stages 10-18 to stabilize neural-domain geometry, lineage allocation, and reporter dynamics; acute loss will reduce the predicted molecular readout and increase pattern variance, while stage-matched wild-type rescue will restore both.

EXPERIMENTAL DESIGN

Use *Xenopus tropicalis* for F0 genetics and *Xenopus laevis* explants. Target neural plate and neural-border lineages by injecting two independent CRISPR RNPs at the 4- to 8-cell stage with a fluorescent lineage tracer; analyze stages 10-18. Groups: uninjected, non-targeting RNP, each guide alone, pooled guides, dose series, wild-type mRNA rescue, perturbation-resistant rescue, and temporally delayed perturbation where a validated small molecule or inducible construct exists. Primary readouts: neural-domain geometry, lineage allocation, and reporter dynamics. Secondary readouts: marker expression, cell trajectories, and neural-tube morphology, amplicon sequencing, RT-qPCR or in situ hybridization, and an orthogonal biochemical or reporter assay. Use at least three independent spawnings, 20-30 embryos per group per spawning for morphology, and 8-10 pooled or individually profiled samples per molecular condition; randomize embryos, blind image scoring, and analyze clutch as a random effect with multiplicity-controlled planned contrasts. Decision point: advance to epistasis and spatial single-cell profiling only if both guides reproduce the effect and rescue restores at least 50% of the molecular and developmental phenotype.

EXPECTED RESULTS AND INTERPRETATIONS

Support would be a guide-concordant loss of the predicted molecular readout, increased spatial or embryo-to-embryo variance, and rescue of both molecular and tissue phenotypes. This would place the mechanism upstream of developmental robustness. A molecular phenotype without morphogenesis would suggest buffering or a parallel tissue effector and would redirect the next aim to sensitized or explant conditions. A morphogenetic phenotype without the predicted reporter change would argue for an alternative biochemical function. No reproducible phenotype after confirmed editing would refute an obligate early role and shift the model toward redundancy, later function, or species-specific deployment.

POTENTIAL PITFALLS

- F0 mosaicism may weaken or spatially blur phenotypes.
- Paralog redundancy or maternal stores may mask an early requirement.

- High-dose perturbation may cause nonspecific toxicity or developmental delay.
- The featured mechanism may not be conserved or may use different effectors in amphibians.
- Abstract-only evidence may omit critical context needed to choose the most specific assay.

ALTERNATIVE STRATEGIES

- Map editing per embryo and enrich analysis for high-editing specimens; establish stable alleles if mosaicism remains limiting.
- Target expressed paralogs jointly, use maternal-zygotic strategies where feasible, or test dominant-negative and degron-based acute perturbations.
- Use lower doses, later induction, tissue-restricted injection, explants, and matched survival/staging controls to separate mechanism from toxicity.
- Test *Xenopus laevis* and *X. tropicalis* orthologs, then validate the key interaction biochemically or in a complementary cell model.
- Before full-scale work, manually review the linked paper's figures and methods and revise the reporter or perturbation to match the demonstrated mechanism.

PUBLISHED ENABLING EVIDENCE AND PROPOSED PRELIMINARY-DATA PLAN

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NOVELTY medium-high - transfers a recent mechanism into a spatially and temporally tractable vertebrate embryo	FEASIBILITY high for the pilots - injection, explants, imaging, genotyping, and rescue are routine <i>Xenopus</i> capabilities	RISK medium - conservation, maternal contribution, and mosaicism are explicit decision points
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GRANT FIT

Exploratory Aim 1 for conservation and causality; successful pilots support Aim 2 epistasis and Aim 3 spatial mechanism.

STRONGEST REVIEWER-FACING PREMISE

Xenopus can test when, where, and whether a newly reported cellular mechanism controls intact vertebrate development in one experimentally integrated system.

SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\) 10.1038/s41586-026-10877-x](https://doi.org/10.1038/s41586-026-10877-x)

Stimulation modulates gene-linked cell assemblies in the human brain

Haley Moore, Mantre Dehnad, Anne Freelin, Bryan Granger, Suganya Subramanian, Tjitse van der Molen, Ashwinikumar Kulkarni, Stefano Berto, et al.

journal-article | Nature 657(8132) | 2026-09-17

DOI 10.1038/s41586-026-10879-9

Access: Abstract retrieved from Europe PMC; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	We found that stimulation strengthens cell assemblies and then linked this effect to cell-type-specific gene regulatory networks.
WHY IT MATTERS	The study advances mechanistic understanding of Stimulation modulates gene-linked cell assemblies in the human brain and creates a defined basis for testing conservation and developmental function in a vertebrate embryo.
MAIN / NEW METHODS	Single-nucleus multi-omics
KEY EVIDENCE	- We found that stimulation strengthens cell assemblies and then linked this effect to cell-type-specific gene regulatory networks. - We further demonstrated the generalizability of these findings by identifying common cell-type-specific gene expression signatures in the human cortex following in vivo stimulation. - Together, our results establish a foundation for identifying targetable genetic signatures linked with physiology that may be harnessed for therapeutic benefit via neuromodulation strategies.
LIMITATIONS	- This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. - The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. - The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of Stimulation modulates gene-linked cell assemblies in the human brain

IMPORTANCE / SIGNIFICANCE

Gene Regulation And Chromatin is central to how embryonic cells convert molecular state into reproducible tissue organization. Establishing whether the featured mechanism operates during early vertebrate development could reveal a conserved control point and a tractable route to explain developmental variability without overstating clinical translation.

KNOWLEDGE GAP

The featured paper does not establish whether the focal regulator or mechanism described by the featured paper acts during early vertebrate development, whether its effect is cell-autonomous, which developmental window is sensitive, or whether molecular and morphogenetic phenotypes can be rescued independently.

RATIONALE AND XENOPUS ADVANTAGE

The paper supplies a candidate mechanism, while early *Xenopus* provides rapid external development, targeted blastomere injection, explants, lineage tracing, live imaging, inexpensive biological replication across clutches, and direct comparison of molecular and tissue-scale outcomes.

CENTRAL HYPOTHESIS

Speculative: the focal regulator or mechanism described by the featured paper acts cell-autonomously during stages 8-14 to stabilize pathway activity, lineage allocation, and spatial marker-domain size; acute loss will reduce the predicted molecular readout and increase pattern variance, while stage-matched wild-type rescue will restore both.

EXPERIMENTAL DESIGN

Use *Xenopus tropicalis* for F0 genetics and *Xenopus laevis* explants. Target animal-cap, organizer, and mesendodermal lineages by injecting two independent CRISPR RNPs at the 4- to 8-cell stage with a fluorescent lineage tracer; analyze stages 8-14. Groups: uninjected, non-targeting RNP, each guide alone, pooled guides, dose series, wild-type mRNA rescue, perturbation-resistant rescue, and temporally delayed perturbation where a validated small molecule or inducible construct exists. Primary readouts: pathway activity, lineage allocation, and spatial marker-domain size. Secondary readouts: cell behavior, developmental timing, and gastrulation morphology, amplicon sequencing, RT-qPCR or in situ hybridization, and an orthogonal biochemical or reporter assay. Use at least three independent spawnings, 20-30 embryos per group per spawning for morphology, and 8-10 pooled or individually profiled samples per molecular condition; randomize embryos, blind image scoring, and analyze clutch as a random effect with multiplicity-controlled planned contrasts. Decision point: advance to epistasis and spatial single-cell profiling only if both guides reproduce the effect and rescue restores at least 50% of the molecular and developmental phenotype.

EXPECTED RESULTS AND INTERPRETATIONS

Support would be a guide-concordant loss of the predicted molecular readout, increased spatial or embryo-to-embryo variance, and rescue of both molecular and tissue phenotypes. This would place the mechanism upstream of developmental robustness. A molecular phenotype without morphogenesis would suggest buffering or a parallel tissue effector and would redirect the next aim to sensitized or explant conditions. A morphogenetic phenotype without the predicted reporter change would argue for an alternative biochemical function. No reproducible phenotype after confirmed editing would refute an obligate early role and shift the model toward redundancy, later function, or species-specific deployment.

POTENTIAL PITFALLS

- F0 mosaicism may weaken or spatially blur phenotypes.
- Paralog redundancy or maternal stores may mask an early requirement.

- High-dose perturbation may cause nonspecific toxicity or developmental delay.
- The featured mechanism may not be conserved or may use different effectors in amphibians.
- Abstract-only evidence may omit critical context needed to choose the most specific assay.

ALTERNATIVE STRATEGIES

- Map editing per embryo and enrich analysis for high-editing specimens; establish stable alleles if mosaicism remains limiting.
- Target expressed paralogs jointly, use maternal-zygotic strategies where feasible, or test dominant-negative and degron-based acute perturbations.
- Use lower doses, later induction, tissue-restricted injection, explants, and matched survival/staging controls to separate mechanism from toxicity.
- Test *Xenopus laevis* and *X. tropicalis* orthologs, then validate the key interaction biochemically or in a complementary cell model.
- Before full-scale work, manually review the linked paper's figures and methods and revise the reporter or perturbation to match the demonstrated mechanism.

PUBLISHED ENABLING EVIDENCE AND PROPOSED PRELIMINARY-DATA PLAN

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NOVELTY medium-high - transfers a recent mechanism into a spatially and temporally tractable vertebrate embryo	FEASIBILITY high for the pilots - injection, explants, imaging, genotyping, and rescue are routine <i>Xenopus</i> capabilities	RISK medium - conservation, maternal contribution, and mosaicism are explicit decision points
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GRANT FIT

Exploratory Aim 1 for conservation and causality; successful pilots support Aim 2 epistasis and Aim 3 spatial mechanism.

STRONGEST REVIEWER-FACING PREMISE

Xenopus can test when, where, and whether a newly reported cellular mechanism controls intact vertebrate development in one experimentally integrated system.

SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\) 10.1038/s41586-026-10879-9](#)

The Virtual Tissues foundation model resolves spatial proteomics across scales

Johann Wenckstern, Eeshaan Jain, Benedikt von Querfurth, Yexiang Cheng, Kiril Vasilev, Matteo Pariset, Phil F. Cheng, Petros Liakopoulos, et al.

journal-article | Nature 657(8132) | 2026-09-17

DOI 10.1038/s41586-026-10884-y

Access: Publisher-deposited abstract retrieved through Crossref; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	In triple-negative breast cancer, VirTues-derived biomarkers predict anti-PD-L1 chemo-immunotherapy response 2 and stratify disease-free survival in an independent cohort 3 , outperforming state-of-the-art biomarkers derived from the same datasets and current clinical stratification schemes.
WHY IT MATTERS	Abstract Spatial proteomics technologies have transformed our understanding of complex tissue architecture in cancer but present unique challenges for computational analysis 1 .
MAIN / NEW METHODS	• Proteomics or mass spectrometry • Quantitative imaging
KEY EVIDENCE	• Each study uses a different marker panel and protocol, and most methods are tailored to single cohorts, which limits knowledge transfer and robust biomarker discovery. • Here we present Virtual Tissues (VirTues), a general-purpose foundation model for spatial proteomics that learns marker-aware, multi-scale representations of proteins, cells, niches and tissues directly from multiplex imaging data. • From a single pretrained backbone, VirTues supports marker reconstruction, cell segmentation and typing, niche annotation, spatial biomarker discovery and patient stratification, including zero-shot annotation across heterogeneous panels and datasets.
LIMITATIONS	• This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. • The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. • The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of The Virtual Tissues foundation model resolves spatial proteomics across scales

IMPORTANCE / SIGNIFICANCE

Cell Mechanics And Morphogenesis is central to how embryonic cells convert molecular state into reproducible tissue organization. Establishing whether the featured mechanism operates during early vertebrate development could reveal a conserved control point and a tractable route to explain developmental variability without overstating clinical translation.

KNOWLEDGE GAP

The featured paper does not establish whether the focal regulator or mechanism described by the featured paper acts during early vertebrate development, whether its effect is cell-autonomous, which developmental window is sensitive, or whether molecular and morphogenetic phenotypes can be rescued independently.

RATIONALE AND XENOPUS ADVANTAGE

The paper supplies a candidate mechanism, while early *Xenopus* provides rapid external development, targeted blastomere injection, explants, lineage tracing, live imaging, inexpensive biological replication across clutches, and direct comparison of molecular and tissue-scale outcomes.

CENTRAL HYPOTHESIS

Speculative: the focal regulator or mechanism described by the featured paper acts cell-autonomously during stages 10-13 to stabilize cell polarity, convergence-extension, tissue strain, and migration persistence; acute loss will reduce the predicted molecular readout and increase pattern variance, while stage-matched wild-type rescue will restore both.

EXPERIMENTAL DESIGN

Use *Xenopus tropicalis* for F0 genetics and *Xenopus laevis* explants. Target dorsal mesoderm and involuting marginal-zone explants by injecting two independent CRISPR RNPs at the 4- to 8-cell stage with a fluorescent lineage tracer; analyze stages 10-13. Groups: uninjected, non-targeting RNP, each guide alone, pooled guides, dose series, wild-type mRNA rescue, perturbation-resistant rescue, and temporally delayed perturbation where a validated small molecule or inducible construct exists. Primary readouts: cell polarity, convergence-extension, tissue strain, and migration persistence. Secondary readouts: fate markers, adhesion dynamics, and whole-embryo axis elongation, amplicon sequencing, RT-qPCR or in situ hybridization, and an orthogonal biochemical or reporter assay. Use at least three independent spawnings, 20-30 embryos per group per spawning for morphology, and 8-10 pooled or individually profiled samples per molecular condition; randomize embryos, blind image scoring, and analyze clutch as a random effect with multiplicity-controlled planned contrasts. Decision point: advance to epistasis and spatial single-cell profiling only if both guides reproduce the effect and rescue restores at least 50% of the molecular and developmental phenotype.

EXPECTED RESULTS AND INTERPRETATIONS

Support would be a guide-concordant loss of the predicted molecular readout, increased spatial or embryo-to-embryo variance, and rescue of both molecular and tissue phenotypes. This would place the mechanism upstream of developmental robustness. A molecular phenotype without morphogenesis would suggest buffering or a parallel tissue effector and would redirect the next aim to sensitized or explant conditions. A morphogenetic phenotype without the predicted reporter change would argue for an alternative biochemical function. No reproducible phenotype after confirmed editing would refute an obligate early role and shift the model toward redundancy, later function, or species-specific deployment.

POTENTIAL PITFALLS

- F0 mosaicism may weaken or spatially blur phenotypes.
- Paralog redundancy or maternal stores may mask an early requirement.

- High-dose perturbation may cause nonspecific toxicity or developmental delay.
- The featured mechanism may not be conserved or may use different effectors in amphibians.
- Abstract-only evidence may omit critical context needed to choose the most specific assay.

ALTERNATIVE STRATEGIES

- Map editing per embryo and enrich analysis for high-editing specimens; establish stable alleles if mosaicism remains limiting.
- Target expressed paralogs jointly, use maternal-zygotic strategies where feasible, or test dominant-negative and degron-based acute perturbations.
- Use lower doses, later induction, tissue-restricted injection, explants, and matched survival/staging controls to separate mechanism from toxicity.
- Test *Xenopus laevis* and *X. tropicalis* orthologs, then validate the key interaction biochemically or in a complementary cell model.
- Before full-scale work, manually review the linked paper's figures and methods and revise the reporter or perturbation to match the demonstrated mechanism.

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NOVELTY medium-high - transfers a recent mechanism into a spatially and temporally tractable vertebrate embryo	FEASIBILITY high for the pilots - injection, explants, imaging, genotyping, and rescue are routine <i>Xenopus</i> capabilities	RISK medium - conservation, maternal contribution, and mosaicism are explicit decision points
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GRANT FIT

Exploratory Aim 1 for conservation and causality; successful pilots support Aim 2 epistasis and Aim 3 spatial mechanism.

STRONGEST REVIEWER-FACING PREMISE

Xenopus can test when, where, and whether a newly reported cellular mechanism controls intact vertebrate development in one experimentally integrated system.

SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\) 10.1038/s41586-026-10884-y](https://doi.org/10.1038/s41586-026-10884-y)

Molecular-level observation of the self-assembly of a virus-like particle

Roi Asor, Dan Loewenthal, Diana Melnyk, Tiong Kit Tan, Philipp Kukura

journal-article | Nature 657(8132) | 2026-09-17

DOI 10.1038/s41586-026-10948-z

Access: Publisher-deposited abstract retrieved through Crossref; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	We show that weak and reversible multivalent interactions control the assembly process by facilitating stochastic selection of a limited set of on-path, topologically closed intermediate structures.
WHY IT MATTERS	More broadly, our approach provides a general framework for visualizing and quantifying the dynamics of multimeric biological machines at the molecular level.
MAIN / NEW METHODS	• Publisher-deposited title/abstract analysis; consult the linked paper for complete technical details
KEY EVIDENCE	• Revealing its underlying principles is essential for understanding biological function 1,2 and malfunction in disease 3,4 . • Viral capsid assembly is the archetypal self-assembly system 5-7 , which has been central in establishing the fundamental principles underpinning biomolecular assembly and the development of new biomaterials 8-10 and therapeutics 11,12 . • We show that weak and reversible multivalent interactions control the assembly process by facilitating stochastic selection of a limited set of on-path, topologically closed intermediate structures.
LIMITATIONS	• This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. • The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. • The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of Molecular-level observation of the self-assembly of a virus-like particle

IMPORTANCE / SIGNIFICANCE

Immunity And Host Response is central to how embryonic cells convert molecular state into reproducible tissue organization. Establishing whether the featured mechanism operates during early vertebrate development could reveal a conserved control point and a tractable route to explain developmental variability without overstating clinical translation.

KNOWLEDGE GAP

The featured paper does not establish whether the focal regulator or mechanism described by the featured paper acts during early vertebrate development, whether its effect is cell-autonomous, which developmental window is sensitive, or whether molecular and morphogenetic phenotypes can be rescued independently.

RATIONALE AND XENOPUS ADVANTAGE

The paper supplies a candidate mechanism, while early *Xenopus* provides rapid external development, targeted blastomere injection, explants, lineage tracing, live imaging, inexpensive biological replication across clutches, and direct comparison of molecular and tissue-scale outcomes.

CENTRAL HYPOTHESIS

Speculative: the focal regulator or mechanism described by the featured paper acts cell-autonomously during stages 8-18 to stabilize cell-stress signaling, epithelial integrity, and lineage-specific transcription; acute loss will reduce the predicted molecular readout and increase pattern variance, while stage-matched wild-type rescue will restore both.

EXPERIMENTAL DESIGN

Use *Xenopus tropicalis* for F0 genetics and *Xenopus laevis* explants. Target epidermal and mesendodermal lineages before mature immunity by injecting two independent CRISPR RNPs at the 4- to 8-cell stage with a fluorescent lineage tracer; analyze stages 8-18. Groups: uninjected, non-targeting RNP, each guide alone, pooled guides, dose series, wild-type mRNA rescue, perturbation-resistant rescue, and temporally delayed perturbation where a validated small molecule or inducible construct exists. Primary readouts: cell-stress signaling, epithelial integrity, and lineage-specific transcription. Secondary readouts: survival, barrier morphology, and developmental timing, amplicon sequencing, RT-qPCR or in situ hybridization, and an orthogonal biochemical or reporter assay. Use at least three independent spawnings, 20-30 embryos per group per spawning for morphology, and 8-10 pooled or individually profiled samples per molecular condition; randomize embryos, blind image scoring, and analyze clutch as a random effect with multiplicity-controlled planned contrasts. Decision point: advance to epistasis and spatial single-cell profiling only if both guides reproduce the effect and rescue restores at least 50% of the molecular and developmental phenotype.

EXPECTED RESULTS AND INTERPRETATIONS

Support would be a guide-concordant loss of the predicted molecular readout, increased spatial or embryo-to-embryo variance, and rescue of both molecular and tissue phenotypes. This would place the mechanism upstream of developmental robustness. A molecular phenotype without morphogenesis would suggest buffering or a parallel tissue effector and would redirect the next aim to sensitized or explant conditions. A morphogenetic phenotype without the predicted reporter change would argue for an alternative biochemical function. No reproducible phenotype after confirmed editing would refute an obligate early role and shift the model toward redundancy, later function, or species-specific deployment.

POTENTIAL PITFALLS

- F0 mosaicism may weaken or spatially blur phenotypes.
- Paralog redundancy or maternal stores may mask an early requirement.

- High-dose perturbation may cause nonspecific toxicity or developmental delay.
- The featured mechanism may not be conserved or may use different effectors in amphibians.
- Abstract-only evidence may omit critical context needed to choose the most specific assay.

ALTERNATIVE STRATEGIES

- Map editing per embryo and enrich analysis for high-editing specimens; establish stable alleles if mosaicism remains limiting.
- Target expressed paralogs jointly, use maternal-zygotic strategies where feasible, or test dominant-negative and degron-based acute perturbations.
- Use lower doses, later induction, tissue-restricted injection, explants, and matched survival/staging controls to separate mechanism from toxicity.
- Test *Xenopus laevis* and *X. tropicalis* orthologs, then validate the key interaction biochemically or in a complementary cell model.
- Before full-scale work, manually review the linked paper's figures and methods and revise the reporter or perturbation to match the demonstrated mechanism.

PUBLISHED ENABLING EVIDENCE AND PROPOSED PRELIMINARY-DATA PLAN

Published enabling evidence and proposed preliminary-data plan: no unpublished user-laboratory data are assumed. The featured paper is the cited cross-species enabling source; direct *Xenopus* evidence was not automatically verified. Pilot 1: map candidate expression or pathway activity at 5-6 early stages using RT-qPCR and in situ hybridization (three spawnings; expected Figure 1A-B); go if a reproducible spatial or temporal signal exceeds twofold over the lowest stage. Pilot 2: test two CRISPR RNPs plus non-targeting control in three spawnings with at least 25 embryos per group, amplicon sequencing, blinded morphology, and the primary molecular readout (Figure 1C-E); go if both guides produce the same directional effect of at least 20% with acceptable survival. Pilot 3: perform wild-type mRNA rescue in three spawnings with at least 20 embryos per group (Figure 1F-G); go if rescue reverses at least 50% of the molecular and phenotype effect. Failure at any threshold triggers the paired alternative strategy rather than a claim of preliminary support.

NOVELTY medium-high - transfers a recent mechanism into a spatially and temporally tractable vertebrate embryo	FEASIBILITY high for the pilots - injection, explants, imaging, genotyping, and rescue are routine <i>Xenopus</i> capabilities	RISK medium - conservation, maternal contribution, and mosaicism are explicit decision points
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GRANT FIT

Exploratory Aim 1 for conservation and causality; successful pilots support Aim 2 epistasis and Aim 3 spatial mechanism.

STRONGEST REVIEWER-FACING PREMISE

Xenopus can test when, where, and whether a newly reported cellular mechanism controls intact vertebrate development in one experimentally integrated system.

SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\) 10.1038/s41586-026-10948-z](#)

TM184C is a GPCR-like regulator of intercellular exchange and autophagy

Kyutae D. Lee, Samuel Taylor, Jennifer Arcuri, Shraddha Chandthakuri, Jacinda Pujols, Bruno Colon, Qiang Wang, Chenzhou Wu, et al.

journal-article | Nature 657(8132) | 2026-09-17

DOI 10.1038/s41586-026-10993-8

Access: Abstract retrieved from Europe PMC; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	By mining millions of AlphaFold2 models 4,5 , we identify two human families of 'superdark' seven-transmembrane proteins-TM184 and PRRT-with structural homology to G-protein-coupled receptors (GPCRs).
WHY IT MATTERS	The study advances mechanistic understanding of TM184C is a GPCR-like regulator of intercellular exchange and autophagy and creates a defined basis for testing conservation and developmental function in a vertebrate embryo.
MAIN / NEW METHODS	• Publisher-deposited title/abstract analysis; consult the linked paper for complete technical details
KEY EVIDENCE	• By mining millions of AlphaFold2 models 4,5 , we identify two human families of 'superdark' seven-transmembrane proteins-TM184 and PRRT-with structural homology to G-protein-coupled receptors (GPCRs). • These bridges mediate organelle sharing through a process that requires the TM184C C-terminal tail and its arrestin code motif 6 , linking GPCR-like beta-arrestin and GRK regulation to vesicle function and intercellular connectivity. • Together, these findings illustrate how structure-based protein discovery can illuminate the dark proteome and identify TM184C as an ancient GPCR-like regulator of autophagy, intercellular connectivity and material exchange.
LIMITATIONS	• This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. • The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. • The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of TM184C is a GPCR-like regulator of intercellular exchange and autophagy

IMPORTANCE / SIGNIFICANCE

Metabolism And Organelle Function is central to how embryonic cells convert molecular state into reproducible tissue organization. Establishing whether the featured mechanism operates during early vertebrate development could reveal a conserved control point and a tractable route to explain developmental variability without overstating clinical translation.

KNOWLEDGE GAP

The featured paper does not establish whether the focal regulator or mechanism described by the featured paper acts during early vertebrate development, whether its effect is cell-autonomous, which developmental window is sensitive, or whether molecular and morphogenetic phenotypes can be rescued independently.

RATIONALE AND XENOPUS ADVANTAGE

The paper supplies a candidate mechanism, while early *Xenopus* provides rapid external development, targeted blastomere injection, explants, lineage tracing, live imaging, inexpensive biological replication across clutches, and direct comparison of molecular and tissue-scale outcomes.

CENTRAL HYPOTHESIS

Speculative: the focal regulator or mechanism described by the featured paper acts cell-autonomously during stages 8-13 to stabilize organelle state, metabolite-sensitive reporters, and lineage-marker induction; acute loss will reduce the predicted molecular readout and increase pattern variance, while stage-matched wild-type rescue will restore both.

EXPERIMENTAL DESIGN

Use *Xenopus tropicalis* for F0 genetics and *Xenopus laevis* explants. Target animal-cap and marginal-zone explants by injecting two independent CRISPR RNPs at the 4- to 8-cell stage with a fluorescent lineage tracer; analyze stages 8-13. Groups: uninjected, non-targeting RNP, each guide alone, pooled guides, dose series, wild-type mRNA rescue, perturbation-resistant rescue, and temporally delayed perturbation where a validated small molecule or inducible construct exists. Primary readouts: organelle state, metabolite-sensitive reporters, and lineage-marker induction. Secondary readouts: survival, developmental timing, and gastrulation morphology, amplicon sequencing, RT-qPCR or in situ hybridization, and an orthogonal biochemical or reporter assay. Use at least three independent spawnings, 20-30 embryos per group per spawning for morphology, and 8-10 pooled or individually profiled samples per molecular condition; randomize embryos, blind image scoring, and analyze clutch as a random effect with multiplicity-controlled planned contrasts. Decision point: advance to epistasis and spatial single-cell profiling only if both guides reproduce the effect and rescue restores at least 50% of the molecular and developmental phenotype.

EXPECTED RESULTS AND INTERPRETATIONS

Support would be a guide-concordant loss of the predicted molecular readout, increased spatial or embryo-to-embryo variance, and rescue of both molecular and tissue phenotypes. This would place the mechanism upstream of developmental robustness. A molecular phenotype without morphogenesis would suggest buffering or a parallel tissue effector and would redirect the next aim to sensitized or explant conditions. A morphogenetic phenotype without the predicted reporter change would argue for an alternative biochemical function. No reproducible phenotype after confirmed editing would refute an obligate early role and shift the model toward redundancy, later function, or species-specific deployment.

POTENTIAL PITFALLS

- F0 mosaicism may weaken or spatially blur phenotypes.
- Paralog redundancy or maternal stores may mask an early requirement.

- High-dose perturbation may cause nonspecific toxicity or developmental delay.
- The featured mechanism may not be conserved or may use different effectors in amphibians.
- Abstract-only evidence may omit critical context needed to choose the most specific assay.

ALTERNATIVE STRATEGIES

- Map editing per embryo and enrich analysis for high-editing specimens; establish stable alleles if mosaicism remains limiting.
- Target expressed paralogs jointly, use maternal-zygotic strategies where feasible, or test dominant-negative and degron-based acute perturbations.
- Use lower doses, later induction, tissue-restricted injection, explants, and matched survival/staging controls to separate mechanism from toxicity.
- Test *Xenopus laevis* and *X. tropicalis* orthologs, then validate the key interaction biochemically or in a complementary cell model.
- Before full-scale work, manually review the linked paper's figures and methods and revise the reporter or perturbation to match the demonstrated mechanism.

PUBLISHED ENABLING EVIDENCE AND PROPOSED PRELIMINARY-DATA PLAN

Published enabling evidence and proposed preliminary-data plan: no unpublished user-laboratory data are assumed. The featured paper is the cited cross-species enabling source; direct *Xenopus* evidence was not automatically verified. Pilot 1: map candidate expression or pathway activity at 5-6 early stages using RT-qPCR and in situ hybridization (three spawnings; expected Figure 1A-B); go if a reproducible spatial or temporal signal exceeds twofold over the lowest stage. Pilot 2: test two CRISPR RNPs plus non-targeting control in three spawnings with at least 25 embryos per group, amplicon sequencing, blinded morphology, and the primary molecular readout (Figure 1C-E); go if both guides produce the same directional effect of at least 20% with acceptable survival. Pilot 3: perform wild-type mRNA rescue in three spawnings with at least 20 embryos per group (Figure 1F-G); go if rescue reverses at least 50% of the molecular and phenotype effect. Failure at any threshold triggers the paired alternative strategy rather than a claim of preliminary support.

NOVELTY medium-high - transfers a recent mechanism into a spatially and temporally tractable vertebrate embryo	FEASIBILITY high for the pilots - injection, explants, imaging, genotyping, and rescue are routine <i>Xenopus</i> capabilities	RISK medium - conservation, maternal contribution, and mosaicism are explicit decision points
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GRANT FIT

Exploratory Aim 1 for conservation and causality; successful pilots support Aim 2 epistasis and Aim 3 spatial mechanism.

STRONGEST REVIEWER-FACING PREMISE

Xenopus can test when, where, and whether a newly reported cellular mechanism controls intact vertebrate development in one experimentally integrated system.

SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\) 10.1038/s41586-026-10993-8](#)

Anodic Pd membrane H2 extraction enhances thermochemical dehydrogenation

Rui Zeng, Julian Ufert, Bryan Y. Tang, Ryan P. Bisbey, Yogesh Surendranath

journal-article Nature 657(8132) 2026-09-17	
DOI 10.1038/s41586-026-11008-2	
Access: Abstract retrieved from Europe PMC; no full-text review is claimed. Evidence status: abstract-supported	
MAIN DISCOVERY	We demonstrate that low anode potentials of 2 from 0.05 atm (balance Ar) to a pure 1.0 atm H 2 stream.
WHY IT MATTERS	This work provides a proof-of-concept demonstration for electrochemically assisted hydrogen removal to enhance selected dehydrogenation reactions.
MAIN / NEW METHODS	• Publisher-deposited title/abstract analysis; consult the linked paper for complete technical details
KEY EVIDENCE	• We demonstrate that low anode potentials of 2 from 0.05 atm (balance Ar) to a pure 1.0 atm H 2 stream.
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SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\) 10.1038/s41586-026-11008-2](https://doi.org/10.1038/s41586-026-11008-2)

CROSS-PAPER SYNTHESIS

RECURRING MECHANISMS

- gene regulation and chromatin (7 papers)
- immunity and host response (3 papers)
- metabolism and organelle function (2 papers)
- cell mechanics and morphogenesis (2 papers)
- neural development and function (1 paper)

METHODOLOGICAL TRENDS

- CRISPR genome editing (4)
- Genetic perturbation and rescue (4)
- Publisher-deposited title/abstract analysis; consult the linked paper for complete technical details (4)
- Single-nucleus multi-omics (4)
- Organoid or stem-cell model (4)
- Biochemical reconstitution (3)

RANKED XENOPUS OPPORTUNITIES

#1 Developmental timing and tissue autonomy of Aneuploidy selects for the acquisition of driver genes in breast cancer

Significance: high if the focal mechanism is conserved | Novelty: medium-high | Mechanistic Clarity: medium pending candidate-specific assay refinement | Preliminary Support: featured-paper evidence; Xenopus pilots proposed | Xenopus Advantage: high for temporal, spatial, rescue, and explant experiments | Feasibility: high for the three rapid pilots | Follow On Aims: epistasis, spatial profiling, and cross-species validation

#2 Developmental timing and tissue autonomy of Prior therapy defines mutation profiles in childhood cancer at relapse

Significance: high if the focal mechanism is conserved | Novelty: medium-high | Mechanistic Clarity: medium pending candidate-specific assay refinement | Preliminary Support: featured-paper evidence; Xenopus pilots proposed | Xenopus Advantage: high for temporal, spatial, rescue, and explant experiments | Feasibility: high for the three rapid pilots | Follow On Aims: epistasis, spatial profiling, and cross-species validation

#3 Developmental timing and tissue autonomy of A compendium of next-generation patient-derived models for diverse cancers

Significance: high if the focal mechanism is conserved | Novelty: medium-high | Mechanistic Clarity: medium pending candidate-specific assay refinement | Preliminary Support: featured-paper evidence; Xenopus pilots proposed | Xenopus Advantage: high for temporal, spatial, rescue, and explant experiments | Feasibility: high for the three rapid pilots | Follow On Aims: epistasis, spatial profiling, and cross-species validation

#4 Developmental timing and tissue autonomy of A tumour-derived organoid biobank maps cancer gene dependencies

Significance: high if the focal mechanism is conserved | Novelty: medium-high | Mechanistic Clarity: medium pending candidate-specific assay refinement | Preliminary Support: featured-paper evidence; Xenopus pilots proposed | Xenopus Advantage: high for temporal, spatial, rescue, and explant experiments | Feasibility: high for the three rapid pilots | Follow On Aims: epistasis, spatial profiling, and cross-species validation

COHERENT MULTI-AIM PROPOSAL

A coherent multi-aim proposal can unite papers centered on gene regulation and chromatin: Aim 1 establishes conservation and timing, Aim 2 resolves cell autonomy and pathway order, and Aim 3 links molecular activity to tissue-scale robustness.

INDEPENDENT PILOTS

- Developmental timing and tissue autonomy of Prior therapy defines mutation profiles in childhood cancer at relapse
- Developmental timing and tissue autonomy of A compendium of next-generation patient-derived models for diverse cancers
- Developmental timing and tissue autonomy of A tumour-derived organoid biobank maps cancer gene dependencies

SOURCE / METHOD NOTE

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